Package 'ewhorm'

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Title Statistical Considerations for Designing e-WHORM Adaptive Trial

Version 1.2

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Description Data simulation and analysis for the design of e-WHORM trial.

URL https://github.com/MartaBofillRoig/ewhorm_sim

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`doInterim_holm_closed_m3` *Interim: conditional errors for Holm closed testing (m=3)*

Description

Interim: conditional errors for Holm closed testing (m=3)

Usage

```
doInterim_holm_closed_m3(z1, v, alpha = 0.025)
```

Arguments

z1	A numeric vector giving first stage z-values computed from first stage p-values.
v	A numeric vector giving the proportions of pre-planned measurements collected up to the interim analysis.
alpha	significance level

Details

eWHORM simulations

Value

A1 matrix of pCERs and CER for intersections 1,2,3,12,13,23,123; same as with gMCP doInterim;

Author(s)

Sonja Zehetmayer

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pcer_bonf_k — *Partial conditional error for a Bonferroni local test at size k (k = |I| is the size of the intersection hypothesis)*

Description

Partial conditional error for a Bonferroni local test at size k (k = |I| is the size of the intersection hypothesis)

Usage

```
pcer_bonf_k(z1, v, alpha, k)
```

Arguments

z1	A numeric vector giving first stage z-values computed from first stage p-values.
v	A numeric vector giving the proportions of pre-planned measurements collected up to the interim analysis.
alpha	significance level
k	is the size of the intersection hypothesis

Details

eWHORM simulations

Author(s)

Sonja Zehetmayer

`sim_trial_pceind_test` *Simulate data from a multi-arm multi-stage trial with shared control and two initial doses, where an additional dose could be added after the interim analysis; individual observations are simulated*

Description

Function to simulate trial data (2-stages, with dose selection). The analyses are performed using partial conditional error rates.

Usage

```
sim_trial_pceind_test(
  n_arms = 4,
  N1,
  N2,
  mu_0m,
  mu_6m,
  mu_12m,
  sg,
  alpha1,
  alpha = 0.025,
  sel_scen,
  side = T,
  test,
  dropout,
  rr,
  bound
)
```

Arguments

`n_arms`	number of arms (including control)
`N1`	sample size stage 1
`N2`	sample size stage 2
`mu_0m`	Baseline value per arm (vector of length `n_arm`)
`mu_6m`	6-month value per arm (vector of length `n_arm`)
`mu_12m`	12-month value per arm (vector of length `n_arm`)
`sg`	sigma covariance matrix between 6- and 12-month mean differences assumed equal across arms (matrix of dim 2x2)
`alpha1`	significance level for dose selection (futility boundary)
`alpha`	significance level for selected dose vs control comparison
`sel_scen`	choose between two different options in case that in interim analysis low dose is promising, but median dose not: 0: do not continue with low dose or median dose; 1: continue with low and median doses
`side`	TRUE/FALSE referring to the side for 1-side testing (if TRUE then lower = side)

test	defines type of analysis: "l" a hurdle model with baseline values as covariables, "w" Wilcoxon test of differences, and "w1" Wilcoxon test of follow-up values
dropout	dropoutrate, between 0 and 1
rr	responder rate for each dose (vector of length n_arm), which gives the proportion of patients with value 0 at follow-up
bound	lower bound to define total responder in simulation study

Details

eWHORM simulations

Value

A list consisting of pvalues at stage 1, pvalues at stage 2, the decision at stages 1 and 2, the selected dose at stage 1, concordance values, corresponding bias and confidence limits and corresponding values on the multi-armed trials.

Author(s)

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